The World That Nobody Wrote

Part One: how I tried to build a civilization instead of simulating one

ARTIFICIAL CIVILIZATION LABORATORY

The World That Nobody Wrote

Part One · the experiments to October 2026 · the project continues

Written from the experiment records of the Artificial Civilization Laboratory, to October 2026. Every number in this book comes from a record in the project's repository, and every person named in my worlds lived in a seeded world.

This is a living book. It is meant to be revised after every major experiment, and earlier chapters are corrected in the open, never quietly rewritten.

Cover and illustrations: AI-generated images, made for this edition.

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github.com/abu4evere/simulated-world

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Prologue — The Vessel

IN THE TWELFTH year of one world, a person named Reibik sat playing with whatever lay nearby: a piece of hide, a piece of bone, some fibre. Reibik tied them together, and what came out could hold water. Nobody had shown Reibik how. Nobody in that world had ever made such a thing, and in all the other worlds running beside it, nobody else ever would.



Reibik beside fresh water, holding the vessel.

REIBIK LIKED IT and carried it everywhere. Reibik made a second one and showed the children how, and they showed others, and the way of making it passed from hand to hand for more than fifty years after Reibik died, changing a little each time, the way stories do.

For two years of that life, Reibik stood within reach of fresh water, holding a thing that could carry it.

Reibik never once filled it.

Not because Reibik couldn't. We can see what Reibik chose to do instead: gather food, give food away, talk, go looking for people. Filling the vessel was never near the top. In that world, the urge to carry water grows with how much thirst a person has known, and Reibik had always lived beside water. Out on the dry land, meanwhile, other people went thirsty. They had never seen a vessel.

The one holding the vessel was almost never thirsty. The thirsty had no vessel.

Is that small urge the whole reason? The record cannot say. It gives us the puzzle, not the answer.

Reibik never existed. Reibik lived inside a computer, in a world I built to answer one question: if nobody writes civilization into a world, will it ever appear on its own? Everything above happened, and every week of it was written down.

This book is about that question, and about the people who lived inside it. It is also about something harder to see: how easy it is, when you build a world, to slip the answer in without noticing, and how much work it takes to keep your own hands out of it.

Before You Begin

THIS IS A living book. My worlds are still running, and every new experiment adds to the story. When something I believed turns out to be wrong, I don't erase it. I write the correction next to it, so you can see how my thinking changed.

Everything in these pages happened in the record. Every name belongs to someone who lived in one of my worlds. Every failure is kept, because the failures taught me the most.

I wrote this book with AI as a tool, the same way I built the worlds. The paintings are imagined illustrations of the moments the record describes.

PART I

The Dream

CHAPTER 1

Why I Built a World

IT STARTED WITH JARVIS, my idea of a personal AI companion, inspired by Iron Man. I didn't want just another chatbot. I wanted something that knew me, could talk with me naturally, help manage my digital life, and become more useful over time.

But as I explored how to build it, I started asking bigger questions. What does it mean for an AI to truly understand someone? Can a system grow beyond what its creator programmed? And if I wanted AI to develop its own behaviour instead of following a script, how much would I have to decide in advance?

Before long I noticed something about almost every AI tool and simulation I looked at. Even when they looked intelligent, I kept wondering how much of what they did was really their own, and how much had simply been written in beforehand. Making one assistant smarter stopped being the question I cared about most.

I wanted to see what would happen if many people lived together in a world: met each other, shared what they knew, formed friendships and families, and learned from what happened to them. Not a world where I wrote the story, but one where the story came from them.

That is how ACL began: the Artificial Civilization Laboratory. The question changed from "How do I build a smart AI?" to "What kind of world would let intelligence and civilization grow on their own?"

The idea didn't come in one moment. It grew slowly, as I thought about intelligence and how complicated things come to be. Nobody designed a language. Nobody designed a city's markets or the first songs. They grew out of many people living, trying things and passing them on. Maybe simple rules and many lives can produce something far more complicated than anyone planned.

So I decided not to program religion, government, culture, or even the way my people talk to each other. I would build only the world and

the people's nature: hunger, cold, fear, friendship, curiosity. Whatever came after that would have to come from them. The boundary between what I build and what the world makes for itself is the thing that fascinates me most. I write the laws of the world. I don't write its history.

A product is built to solve a problem you already understand. You know what it should do, and you make it do that better. ACL is built for questions I can't answer. Will my people find their own ways to communicate? Will traditions form? Will they help each other, fight, or make things I never imagined? I don't know. That is exactly why I want to find out. It may take a long time, and the answers may not be the ones I hope for. But even a failure can teach me how complicated behaviour grows out of simple things.

There was a second question underneath the first, and I gave it a name: Bob.

Bob is not a character. Bob is a question about one person. If someone were born inside a world like mine, learned from experience, made friends, kept memories, developed fears and habits, would that person be someone? Would they still be themselves if what they lived through were carried somewhere else, into another body, into our world?

Bob is the place I hope ACL leads. If we get there, we won't just have a simulation full of figures following rules. We'll have to face a much harder question: can someone born inside a simulated world truly become someone? And if we ever manage more than one, there will be Bobs.

I don't claim we are there. Nothing in this book shows that anyone in my worlds is conscious or feels anything the way we do. Bob is the destination. ACL is the journey toward finding out whether we can reach it. It is also why every person in my worlds is a particular someone, with a name and a history, never just a number in a crowd: if the question about Bob is ever going to mean anything, the first question has to be answered with people.

If you remember one thing from this book, I want it to be this: you can write the laws of a world without writing its history. But if you want

to know what really grows there, you have to let the evidence argue with you, even about the world you built yourself.

CHAPTER 2

Games That Already Know the Answer

YOU HAVE PROBABLY played a civilization game. You start with a few huts, you research pottery, then bronze, then writing, and a few hundred turns later you have cities, armies and a space programme. It feels like watching history unfold.

It isn't. Somebody decided long before you pressed "start" that bronze comes after copper and writing after pottery. When your little people discover bronze, nothing has been discovered. The game's designer discovered it, years ago, and wrote it on a list. A happiness meter that rises when you build a temple is not religion. A trade screen is not trade. A government picked from a menu is not politics.

These games are often beautiful, and they are honest about what they are: games. They never claimed to be experiments. The trouble starts when we treat them as if they could tell us something about how real civilizations began. If the outcome is written into the rules, the outcome can't prove anything. It is like hiding a coin in your own pocket and then being amazed that you found it.

There is a quieter version of the same trick, and it worried me more, because it is easy to fall into even when you are trying hard not to. Many simulations avoid lists of inventions but still reward the right answer. Their people are paid for inventing, paid for trading, paid for cooperating. Sure enough, inventing, trading and cooperating appear, and it looks like they grew on their own. But the reward was the recipe. The designer just moved the answer from the rules into the reward, where it is harder to see.

Real life doesn't work like that. Nobody paid the first person who tied a stone to a stick. Nobody awarded points for the first song. A good idea in the real world earns nothing until it happens to make someone's life a little easier, and even then most good ideas are forgotten.

So I set my world a harder test, and I have tried to keep it ever since:

- Nothing my people think with may contain a list of inventions.
- Nothing may reward an invention just for being an invention.
- Nothing may tell anyone what is useful.

IF CIVILIZATION APPEARS under those conditions, it means something. If it doesn't appear, that means something too. Either way, I will have learned something real.

CHAPTER 3

What Would Count as Success

BEFORE I COULD build anything, I had to be clear about what I was building, because it is easy to confuse with three other things.

A **game** is built to be played. Its rules exist to make your choices interesting. Whether its history is believable doesn't matter, and shouldn't.

A **simulation of one thing** follows many small actors under simple rules and watches what they do together: birds flocking, cars jamming on a highway, a disease spreading through a town. These are often excellent science. But each studies one thing, and the ingredients of that thing are put in on purpose. A model of traffic has cars and roads in it from the start.

A **simulation of history** models the big pieces: states, economies, wars, technologies. Those big pieces are what the model is made of. People, if they exist at all, are parts inside them.

What I wanted was none of these. My world has no player and studies no single thing. It has no states, markets or religions written into it at all. It has individual people, a physical world, and time. Anything bigger than one person, a family custom, a shared sign, a way of doing things, has to be built by people out of what they do, or it doesn't exist.

That kind of world may not even be possible. Finding out is part of the point.

So what would success look like? The word I kept coming back to was **open-ended**. A world is open-ended if it keeps producing things that are genuinely new: new kinds of objects, new habits, new traditions, not just new arrangements of the same few pieces. Life on Earth is like that. Human culture is like that, so far. Almost no simulation is. Most settle down after a while into a steady state where nothing new ever appears, like a snow globe after you stop shaking it.

I decided from the start to say in advance what would count as success and what would count as failure, so that I could never later call a failure a success just because I wanted it to be one.

Success, at its smallest, would be any culture at all appearing without being scripted: a way of doing things passed from one person to another, a mark that comes to mean something, knowledge that outlives the person who found it. At its biggest, it would be a world where new things never stop coming, where year five hundred holds things that year fifty couldn't have held.

Failure could come in three shapes, and I named each one so I would recognise it:

- **Nothing happens.** People live and die and never build up anything.
- **Something happens, but I put it there.** The result is real, but the experiment is worthless, because the answer was hidden inside.
- **Something happens, then stops.** Culture appears, settles, and freezes.

By October 2026 my worlds had clearly escaped the first failure, fought hard against the second, and walked straight into the third. Most of this book is about the third.

PART II

The Rules I Gave Myself

None of the rules in this part were handed down at the start. Most of them were written after something went wrong. Each one is a scar.

CHAPTER 4

Nature, Not History

THERE IS ONE rule underneath all the others, and every other rule is a way of keeping it: **I write nature. Nobody writes culture.**

I write the physics, the bodies and the feelings: what hunger is, how winter works, how flint breaks, what grief does to a person. I don't write tools, fire-making, shelters, farming, trade, leaders, feuds or myths. Those have to come from people living under nature's rules. In one sentence: I write the laws of the universe, not its history.

It works like the real world. Nobody gave the first humans a recipe for fire. They had bodies that felt cold, hands that could strike stones together, and eyes that noticed sparks. Fire came out of that, eventually. I give my people the bodies, the hands and the eyes, and then I wait.

This matters to me because I don't want to write the history of my civilization. I want to create the conditions in which history can happen. I can decide how much food a person needs, how memory works, how children are born, and how news can travel. Those are the rules of the world. But I don't want to decide what my people will invent, what they will believe, which traditions they will keep, or what kind of society they will build. If I put the answers in, I already know the ending, and I would only be watching my own decisions play back to me.

The obvious shortcut was a small list of inventions, just a few steps to get things started. I rejected it, because there is no safe size for a script. If the first step is written, every later step stands on it.

The rule is brutal in practice. My people start with nothing and may never get anywhere. It means long stretches of watching a world do very little. And it means the line between nature and culture has to be argued case by case. Is the fear of wolves nature or culture? My answer: being able to feel fear is nature. Being afraid of wolves in particular is learned, from being hurt, from seeing someone hurt, or from being told.

Two more rules grew out of this one.

Nothing is heading anywhere. My people have no purpose built into them beyond staying alive and the pulls of their own feelings. Nothing in them wants to invent, to advance or to build. Nothing rewards a world for becoming more complicated. If I nudged my worlds toward complexity, then finding complexity would prove nothing; it would be like tilting a table and being surprised that the marbles roll one way. My people do have curiosity, a pull toward the untried that comes from their temperament, the way some children take every toy apart and others don't. But curiosity is not a reward for being inventive. When people in my first long run made blades in their very first year, nobody had wanted a blade. Someone struck flint against flint, often while playing, and was left holding a sharp edge. The edge helped them gather food. Whether a blade ever appears in a world depends on whether that accident happens, and whether anyone feels the difference.

No invention is special. If my code treated some inventions as special, with their own rules or names, then those inventions were chosen in advance, by me. A rule for "spear" is a step on a hidden list even if no list is ever drawn. So every made thing in my worlds is just an ordinary result of the same physics. The part of the code that decides what people do contains no names of inventions at all, and a test reads the code and fails if words like "pot", "kiln" or "cooking" ever show up there. That test has caught us once already: during one of our big checks, an event had been given one of the forbidden words as a name, the test failed, and the name was changed. Names live only with the historian, a watcher who stands outside the world and, after the fact, calls a thing a raft or a blade by what it can do. Another test makes sure the historian never changes anything a person does.

We have found shortcuts in my worlds, places where people could use information they shouldn't have had. Those didn't make every result false, but they forced me to ask which conclusions my worlds could really support. If ACL only worked because I had secretly put the answers in, I would be disappointed. But I would rather discover that than spend years believing something that wasn't true. When we find a problem, I look at what it affected, keep the results we can still trust, fix the rules

and run the experiments again. I don't erase the old results or pretend the problem never happened.

The goal isn't to make ACL look successful. The goal is to find out what it can really produce. **I want to program the possibilities, not the outcomes. I want to write the laws, not the history. And if the evidence shows I failed, I want to know.**

CHAPTER 5

A World That Doesn't Care

MY WORLD HAS no main character, and it protects no one. Rain falls where it falls. A storm puts out the fires of the careful and the careless alike. A band of five people on a small island, with no partner for anyone, dies out, and nothing steps in to save them.

That is how the real world works, and it's the only honest way to build one. A world that rescues the last survivors, or sends rain just as a people is dying of thirst, is a world with an author writing a happy ending. If I protected my people, every success they had would be partly mine.

The price is extinction. In my first long run, two of eight worlds died. In my thousand-year run, three of eight. In a run of twenty-four worlds a little later, after I had made the seas harder to cross, nine of the twenty-four were gone within two hundred years. These deaths are not bugs. They are what an uncaring world does. My job is to write them down and ask why, not to prevent them.

The world doesn't care, but it is consistent, and that consistency is where everything useful comes from. In my world, materials aren't items on a menu. Each is a bundle of physical qualities: how sharp it is, how long, how hard, how well it holds things, how warm, how well it floats, how tightly it binds, how soft it is. There are two ways to change them: strike one thing against another, or tie two things together. What a made thing can do follows only from its qualities. A sharp edge on a long handle helps with hunting. Something that holds helps carry food. Something warm helps in winter, if you wear it.

Nobody listed the raft. The raft exists because wood floats a little, tying two floating things together floats a bit better, and enough float lets a person stand on water. The first person to tie two sticks together made a raft, whether anyone meant to or not.

That is exactly what happened, and it taught me one of my most important lessons. My worlds start as islands, one founding band on

each, so the peoples stay apart unless someone works out how to cross the water. The first question anyone asked of my first long run was: when will they invent the raft? The answer was almost embarrassing. Immediately. Wood is the commonest thing near any camp, and in every world someone tied two sticks together within the first three years, often within weeks, usually while playing. They had made a boat. Nobody knew.



The raft nobody used.

AND NOBODY USED it. Almost every crossing between islands in that run happened without anything that floats, over narrow channels that my movement rules had quietly let people jump. Across open water, there were three crossings in nearly fifteen hundred years of world-time. In each world, dozens of people knew how to make a raft, and almost no one carried one. When we looked for the reason, we found three: nothing in the world ever showed anyone what floating was for; people carry only the few things they value most, and a raft was valued like a slightly awkward club; and the rare person who did set out to sea with one picked a random far shore, a different one every week.

The raft was invented and forgotten in the same moment.

The fix was not to reward rafts. That would have been writing history. Instead I changed the world. Water now really blocks walking. Far shores can be seen from the coast. And landing on new land became something you feel, the moment it happens, visible to anyone watching from the shore. After that, open-water crossings rose from almost none to anywhere between a handful and more than a hundred and fifty in a single world. There was a price, and it took two more experiments to see it: with the jump gone, many small bands were stranded on their islands, and stranded bands die. The change that let rafts matter made the world harsher for anyone who never built one.

The raft taught me a rule I have kept ever since: **a use has to be lived.** A thing isn't useful because I know what it's for. It becomes useful when someone in the world goes through something that matters to them because of it. Having a raft means nothing. Landing on a new shore with one means everything.

And that led to the question I now ask before adding anything new to my world: am I creating a condition people can use, or am I quietly scripting the result? Don't ask how to create trade; ask what conditions would make swapping things worthwhile. Don't ask how to create government; ask what pressures would make working together valuable. Don't ask how to make people fill vessels with water; ask what in their world could ever show them that a vessel holds water. When something I hope for doesn't appear, the temptation is always to add it. My rule is to add the condition, never the outcome, and if the condition isn't enough, find out why before adding anything else.

CHAPTER 6

Where I Fell Short

A BOOK THAT boasts about its principles should also admit where it breaks them. Here are the places, known by October 2026, where my world falls short of its own rules.

The first is the words. When you watch my world on screen, people seem to talk to each other in English. They don't. What you see is my translation of what really passed between them: a warning, a piece of news, a lesson. My people have no language of their own yet. The marks they scratch into rock and bark are the only symbols that are truly theirs.

The second is a piece of knowledge I gave them by accident. A person carrying something that floats well enough will set out across the water toward land they have seen, and sometimes at random. Knowing that a floating thing is a way across the sea is exactly the kind of knowledge they should have to learn. A code review found it in October 2026, and it hasn't been removed yet.

The third is that the kinds of things that can pull at a person's mind are a fixed list: hunger, worry, a puzzle, grief, cold, and a few more. Goals can attach to particular things a person has found, but the kinds of goal can never grow. A real person can come to care about honour, or beauty, or a god. Mine can't, yet.

The fourth hurt the most when we found it, because it falls on exactly the inventions that cumulative culture is made of. Each person can remember how to make at most twelve things. A new one beyond that pushes out the one they value least, and a brand-new invention starts valued at nothing, so a full memory often loses a thing the moment it is made. Worse, when two recipes are valued equally, the tie is broken by the recipe's name, and the names of things built from other made things happen to sort first. So among equals, the more complicated invention is always the one thrown away. Nobody designed that as part of human

nature. It was an accident of bookkeeping, found in October 2026 while we were counting what people keep.

The fifth is how narrow watching is. In real life, a child learns most things by watching. In my world, a person can work out how something is made by watching someone else in only two situations: seeing them land on a new shore with a raft, and hunting beside them. A carrier, a warm wrap, a blade used for gathering food or a piece of fired clay can be inherited, carried and lived beside for a lifetime without anyone ever learning how it was made.

The last two are about habits. My people learn their habits for a fixed set of situations; they can't invent a new kind of situation to have a habit about. And not all of their habits are learned from their own lives. Some are inherited at birth, as a softened copy of their parents'. Some are soaked up from whoever raises them. Some are picked up in conversation. Only part comes from their own experience. For a while the screen that shows a person's mind said how many of their habits they had "learned", as if every habit came from experience. That was wrong, and in October 2026 I changed it to show only what a habit is, not where it came from.

A history that records its principles but not its failures is just an advertisement. I'd rather you know where the cracks are.

PART III

What Happened

From here on, this is the story of what my worlds actually did, in the order it happened. Before every experiment I wrote down what I expected, and locked it away where I couldn't change it. Many of those guesses were wrong. I've kept every one.

CHAPTER 7

The First Worlds Come Alive

THE VERY FIRST experiment was small and careful: ten people in two bands of five, living for a little over a year. I wanted to know one thing. Would my people do anything you couldn't explain as a simple reflex, like a thermostat switching on when it gets cold? I wrote down my guesses: that survival would come before everything else, that people would get good at predicting what foraging and resting would bring but bad at predicting each other, that friendships would feed themselves, and that the two bands would never meet.

I don't have the scored results of that first run. The guesses survived; the report didn't. I've left that gap as a gap, rather than fill it in from memory. What the first run did leave was a habit I have kept ever since: write the predictions down first, lock them away, and only then press "start".

The second experiment was the first time I let worlds run long enough to have a history. Eight worlds, up to two hundred years each. Twenty founders in every world, split into four bands of five, each band on its own island.

The worlds came alive almost at once.

In one of them, in the very first year, a person named Ruleidon went back to the place where they had left something, for the third time, to see if it would happen again. It was the first deliberate test of anything in that world: someone checking whether the world behaves the same way twice. In year four, a person named Tairo struck flint where dry things lay, and they caught. It was the first fire ever made by people in that world. Nobody had told Tairo that sparks make fire. Tairo had puzzled over sparks, and tried.

In year fourteen, in the same world, a person named Roukarail attacked a person named Kaileir. It was the first violence anyone there had done. It was a hungry year; three people starved after a poor growing

season. Six years later, a person named Ludol attacked Kaileir. Kaileir fought back and killed Ludol. Revenge answered revenge, and a feud began between Kaileir's family and Ludol's.

Then the story of that killing began to travel. It went the only way anything travels in my worlds: parents telling children, neighbours passing on news, each person telling it as they remembered it. Eleven years after the killing, three people said it was someone called Rukeimei who had killed Ludol. Eighty years after, sixteen people believed it was Naikoubo. A hundred and eighty years after, fifty-one people believed it was Midigain. None of them had been there.



A memory passed on by the fire.

AND YET THE memory itself held. A hundred and eighty years after the killing, sixty-two people across five generations still remembered that Ludol had been killed. It was the longest-lived memory in all eight worlds. They had simply lost who did it.

That was the first time I saw my people carry history beyond living memory, which is the beginning of what a culture is. And in the very same story, I saw that what a people remembers is not the same as what

happened. Think of a game of telephone, played across five generations. A people's past is something they make out of retellings. My world doesn't correct them, and shouldn't.

More of what I had hoped for showed up in those first eight worlds. Six of the eight survived. Every one of them developed shared marks, scratched into rock and bark, that came to mean water, shelter or a grave, more than seven hundred of them in all, and about a third of those marks reached people born after they were made. Myths, stories that weren't true but were told by several living people, appeared about twice a century in every world. In every surviving world, someone struck fire after puzzling over sparks. And every world had invented every named kind of thing within a hundred years.

Some of my guesses were badly wrong, and those were the most interesting. Stories went wrong far more often than I expected: in one world, three quarters of what people believed about the past was false. And trust had nothing to do with truth. Whether a person was usually right made no difference to how much others trusted them. People trusted whoever fed them. It is not hard to think of human examples.

Then there was the raft, invented almost at once and almost never used, the story I told in Chapter 5.

So I made four changes, all at once. People would now earn their reputation by being right when it was checked, not just by being generous. They would have many more chances to check what they were told. The traces and marks left behind would be able to teach. And floating would become something felt, with the narrow channels between islands no longer jumpable.

Every change worked. People checked what they were told more than fifteen times as often. Reputation began to follow truth. People crossed open water in every world. In every surviving world, someone learned a technique from a mark another person had left, and in three worlds, a skill that nobody alive still knew came back because somebody read it from a mark.

And people started dying.

After a hundred years, my worlds held about forty percent fewer people than before. I had written down, in advance, that I would call the changes a failure if the population fell below half. It came within five people of that line. Half my predictions were wrong. The causes of death looked the same as before; births had simply fallen by a third. Nobody could say why.

So I stopped building and started digging. I replayed the first century of all sixteen worlds, the eight before the changes and the eight after, and checked that each replay matched the original exactly, the way you might watch a recording of a game frame by frame. Then I followed the path from a woman able to have children to a birth, step by step.

Before looking, I wrote down my guesses with odds. My money, at sixty percent, was on the islands: with the narrow channels closed, fewer people could reach a partner. The other main guess was that people were simply busier, with all the new things to check.

Almost every explanation failed. Women had children at almost exactly the same rate as before. People spent their time almost exactly as before. Babies went hungry about as often as before. One world showed my island explanation perfectly. There, almost nobody had a possible partner on their own island; the share of women with a partner halved, births fell to about a quarter, and the world dwindled away. But in the overall numbers, that one world was hidden among the others.

The honest answer was an uncomfortable one: eight worlds were not enough to tell a real effect from bad luck. It's like tossing a coin eight times and getting six heads. It might be a bent coin. It might just be chance.

So the next experiment was built to settle it, with twenty-four worlds, each run twice: once under the old rules and once under the new. I decided in advance that I would only believe the new rules lowered the population if they did so in at least seventeen of the twenty-four. They did in eighteen. The effect was real.

The cause was my island guess, but it worked in a way I hadn't pictured. Every world starts with four bands of five. Under the old rules, seven bands in ten eventually had a child with another band. Under the

new rules, only four in ten did. And a band of five that never mixes with anyone almost always dies, under either set of rules. Within a few generations everyone on the island is related, and no one is left to pair with. Fewer than one isolated band in five still had a living descendant after a hundred years. The new rules hadn't made isolation deadlier. They had made it far more common: twice as many bands never met another band at all.

It cost more than lives. With fewer people mixing, fewer techniques were tried, and the ones that were tried were forgotten sooner. Fewer people keep fewer things alive.

I kept the four changes. Their good effects held in every world that lived. But I had learned two lessons I would meet again and again. A wall of water is a death sentence for a small people. And any change to the physics of a world can reach all the way into who gets born.

CHAPTER 8

A Thousand Years, and the Freeze

THE EXPERIMENT I had been waiting for was the long one. Eight new worlds, every rule locked, a thousand years each. I expected time to give culture room to grow. If a people had a thousand years, surely they would build something on top of what they started with.

My written predictions were cautious, and most of them came true. Three worlds died: two within their first fifty-five years, from hunger, thirst and children left uncared for, and one slowly, after four centuries, mostly from illness. Five reached year one thousand alive, each with somewhere between thirty-six and eighty-six people. For nine centuries their populations held almost flat, births matching deaths, like a village that never grows and never empties.

In almost every world, one people remained, and usually not a mixed one. In four worlds, everyone alive at year one hundred descended from a single founding band of five. The other three bands had left no one. The usual path to one people was not merging. It was one band outliving the rest.

One world proved me wrong in the most beautiful way. I had predicted there would be no separate peoples. In that world, two islands lay side by side for nine hundred and ninety years, each with between fifteen and fifty-three people. In a thousand years, not one person on either island formed a relationship of any kind with anyone on the other. They never met.

Their temperaments drifted apart early and stayed apart, while the way they differed kept changing: at first they differed most in how hard they worked, centuries later in how anxious they were and how much they cared for others. Their cultures grew apart in ways nobody designed. One island kept beliefs about leaving fibre, stone and wood at certain places in certain seasons, an old building practice. The other island never held any of those beliefs, in any year I checked, for a thousand years. The

second island bound things with hide and worked bone; the first did neither. Their marks shared nothing: each island had made its own shapes for rabbits, for graves and for water. They were two peoples with two histories, across a stretch of sea, and neither ever knew the other existed. Think of two valleys in the mountains that grew their own dialects and customs for centuries without ever crossing the ridge between them.



Two peoples, a strait apart, who never met.

SOME THINGS LASTED the whole thousand years. The simplest carrier, fibre tied to fibre, was held in five worlds from year ten to year one thousand. About three in four of the people who came to hold it had been taught by someone. The rest found it again by themselves, because it is the easiest thing in the world to make.

But my predictions hadn't asked the question that mattered most, and the results answered it anyway.

I had expected time to do the work. A thousand years is forty generations. Surely, I thought, somewhere in those forty generations someone would build on what their great-great-grandparents had found.

So when the results came in, the first thing I looked for was the newest thing each world had ever made, and when it had first appeared.

Every kind of tool any of those worlds ever had, whether carrier, boat, club, blade, spear or warm wrap, first appeared within the first eleven years. After that, for nine hundred and ninety years, nothing new appeared. Not one new kind of thing.

Worse, the worlds slowly forgot. The number of ways of making things that people knew peaked somewhere around year fifty to a hundred, and by year one thousand it had fallen to about a tenth. Much of what was lost was the clutter of play. But not all of it. Spears were lost in every long-lived world, sometime between the second and fifth centuries, and never came back. Warm wraps were lost in all of them.

A world at year one thousand looked much like the same world at year thirty, and sometimes poorer.

A freeze that complete could have been a broken counter, so it had to be checked against the worlds themselves. It wasn't a bug. The experiment had worked exactly as designed, and that made it impossible to wave away. These worlds had history. What they didn't have was history that kept going.

The worlds went on. People lived their lives. Generations passed. But culturally, something had stopped moving. I had to face a hard possibility: building a world where people survive, talk and have children does not automatically build a world where culture keeps growing. Maybe my people lacked something they needed to invent. Maybe new ideas appeared and failed to spread. Maybe the groups were too small or too cut off to keep and build on what they learned. Or maybe the world rewarded staying alive without ever rewarding discovery.

I couldn't simply make my people invent more often and call the problem solved. That would be writing the answer in instead of understanding why culture had frozen. So the freeze became something to investigate, not something to hide. My question changed. I was no longer asking only whether a civilization could last a long time. I was asking what lets a civilization keep creating something new.

My first answer was the obvious one: maybe the world was too poor. With six materials and two ways of changing them, perhaps there simply wasn't enough to discover. So I made matter richer. Things now had weight, warmth, wetness, softness. Soft wet clay lay by rivers and lakes. People could hold something in a fire, fill something, and press soft stuff into a shape. I ran eight worlds, each for a thousand years, and predicted that in at least half of them, something new would still be appearing after year three hundred.

It didn't happen. In every world that lived three centuries, the last new thing appeared by year sixty-six. My people did take up the new physics, but only the shallow end of it. They shaped clay constantly, dozens of shapes in every world. And then they stopped. In all eight worlds, over up to a thousand years each, exactly one thing was ever changed by being held in a fire. Nobody ever filled a vessel. No water and no fire was ever carried anywhere. The richer worlds were also harsher, and I never found out why: four of the eight died out.

There was one small hopeful sign. In two of the surviving richer worlds, new traditions began after year one hundred and lasted a century or more; in one, a belief about leaving fibre at certain places in summer and autumn began around year three hundred and forty and lasted to the end of the millennium. No world under the old rules ever had a tradition that began that late. Two worlds against none is a hint at most, but I wrote it down, because it may matter later.

So I stopped building new worlds and started taking old ones apart. I replayed two of the richer worlds from their first week, with instruments that only read and never touch, and recorded every choice every person made, and every option they had weighed before making it. Before trusting a single number, I checked that the replays matched the saved worlds exactly. They did.

That is how I found the two people at the heart of this book.

The first was a child named Beimak, born in the world's eighth year to two of the founders. Beimak was curious, more than most, not very hard-working, not very sociable, quick-tempered and a little anxious. At about four years old, playing at a fire, Beimak held a dry lump of clay in

the flames. The clay set hard. It was the first fired clay in that world, and as it turned out, the only fired clay that world would make in a thousand years: the whole history of pottery in that world.



The only fired clay that world ever made.

BEIMAK REMEMBERED HAVING made it, but valued it at nothing. Beimak never used it, never made it again, and never showed anyone. In the weeks that followed, Beimak's choices were ordinary ones: rest, gather stone, play, talk with an older sibling. Nine weeks after the firing, Beimak's mother starved to death. Three weeks later, Beimak's youngest sibling, not yet three, died of thirst. Two weeks after that, Beimak starved too. In five weeks the family had lost a mother and two children. Two weeks later, the world recorded that nobody alive knew how clay could be made hard in a fire. The knowledge had lasted sixteen weeks.

Beimak showed me that the physics worked: heat did exactly what it should. It showed how rare the chance was: in a whole century, adults stood at a fire holding a piece of clay that could be fired in only three weeks. And it showed something harder to see in tables. In my world,

the people most likely to try something new are the ones who play, and the ones who play are children, who are also the ones most likely to die.

The second was Reibik, the person from the beginning of this book. Reibik was one of the founders of another world, very sociable, calm, not especially bold or curious, with a partner and six children. In the world's twelfth year, playing with materials, Reibik bound hide to bone with fibre and made the only watertight thing ever made in any of those eight worlds.

You already know what Reibik never did with it. What the replays added was the rest of its life. Reibik made a second one, and taught it: to one child, then another, then a third. The children carried the recipe on after their parent was gone. One of them taught it to a ten-year-old outside the family, who later taught it to their own child, and to a half-sibling. By then the recipe had changed in the telling: the later learners believed it was tied with hide, not fibre. The knowledge lived on for sixty years, until Reibik's child Baigi starved, and the next week nobody alive knew it anymore.

When I look at the two years Reibik stood beside fresh water with it, I see a question bigger than the vessel. The invention happened. The ability existed. But the act that would have made the invention matter never followed. Reibik forces me to separate things that look like one thing. Making a vessel is not the same as understanding what it can do. Understanding is not the same as remembering to use it. And remembering is not the same as choosing to act. Reibik's story alone doesn't tell me which of those failed. The record gives me the puzzle, not the explanation. But it showed me that I can't count an invention as the end of the story. I have to follow what happens next: whether it is used, whether someone else learns it, whether it survives, and whether it becomes part of life. **Reibik created the possibility, but the possibility never became a practice.**

The replays let me score every link in that chain, separately for fire and for water. Was there an opportunity? For fire, rarely, because fires are rare and brief. For water, constantly. Did anyone take the first step on purpose? Never, for either; the one fired lump came from a child's

play. Did the material change as it should? Every time. Did anyone notice what it meant? Barely. Did it spread? The clay vanished in sixteen weeks; the vessel was taught for sixty years, but only ever as a carrier.

Before looking, the favourite explanation had been that knowledge was simply lost, that someone discovers a thing and it dies with them. Knowledge loss was everywhere in my worlds, after all. But that wasn't the main problem here: Reibik's vessel was taught to seven people and outlived its maker by decades. The problem was earlier. People met the opportunity and never took the first step. I gave that place a name: **the first-try bottleneck**.

One more suspicion had to be checked. Maybe my people never filled a vessel because the reward comes weeks later and far away, and they simply couldn't learn from delayed rewards. The habit part of their minds does forget fast: a reward five weeks after an act counts only a third as much. But I found two other ways they learn that reach much further back. Beliefs about leaving things at places learn from outcomes up to a year later. And how much a person values a thing they made learns with no time limit at all, as long as they still hold it: things first used more than a year after they were made were made again three times as often as things used at once. If Reibik had ever filled the vessel and drunk from it later, the lesson would have reached Reibik.

So the problem wasn't learning from delay. It was the first step itself, which no delay can teach, because it never happens.

CHAPTER 9

Slowly Forgetting

ONCE I KNEW the problem was the first step, I had to decide which side to mend: the world, or the people.

I could change my people: make them more curious, better at planning, more eager to invent. Or I could change the world: let nature take the first step by itself, the way crossing water once showed the raft its worth. I chose the world. I wanted to understand what could grow from the people I had already made. If they weren't inventing, I wanted to know why. Maybe their world gave them no reason to. Maybe useful knowledge couldn't spread. Maybe their groups were too small. Making them want to invent more might fix what I could see without answering what I wanted to know.

That isn't an absolute rule for me. Changing how my people think can be a fair experiment too, as long as I'm honest about what I changed. If I give them a new ability and their civilization blossoms, that doesn't prove they would have grown that ability themselves. It only shows what becomes possible under the new conditions. My default is to look at the world before rewriting the people. The point is never to protect my original people forever. It's to know which conditions lead to which outcomes, without pretending I discovered something I actually programmed.

So I let nature take the first step. Rain would fill any open watertight thing. Wading into a river or lake would fill it too. A fire hot enough would harden dry clay lying beside it. My people's minds stayed exactly as they were, not one line changed. Then, before any world ran, I went through the new code line by line against what I had promised to test.

I found six problems. One of them would have killed the experiment by itself. When a fire hardened clay, the recipe was stored under the wrong label, and the next time that person made a decision, the whole world would crash. Every test had passed, because none of them let a

world keep running after a firing. The very first piece of fire-hardened clay in any world would have stopped everything, and I would have recorded the experiment as a failure for a reason that had nothing to do with the question. The other five were quieter, but each would have meant testing something other than what I had promised: rivers in my world can be waded, so they never counted as water and never filled anything; everyone standing near a fire had all their food treated as cooked; and so on. I fixed all six, wrote them down separately, and locked the fixes away before the first world ran. An experiment can be lost before it starts, quietly, inside one wrong label.

Then sixteen worlds ran for a thousand years each.

Half of the idea worked. Fires hardened clay in front of people more than six hundred times, where before it had never happened. People kept what the fire made. Twenty-five fired objects outlived the people who first carried them, and some were still being carried fifty years later.

The other half never had a chance. In eight worlds over up to a thousand years, rain or a river filled a vessel three times, and nobody ever drank water they had carried. The world can't fill vessels nobody makes.

And novelty still stopped. Across the eight worlds, more than six hundred lasting new things appeared in the first century, and exactly one after it.

That one late novelty excited me for a while. It appeared in year one hundred and fifteen, in the world where the two peoples never met, and it looked like a world that had almost found something new. If it was real, it was the first sign in all my worlds that a people could still find something new after its first century, the very thing the thousand years had not shown me. That is exactly why it had to be taken apart before anything was built on it. I replayed that world to year one hundred and thirty-five, checked that the replay matched the original at every ten-year mark, and then followed the new thing back to where it came from.

It wasn't new. It was a flint flake. The flake had first been made by a person named Daiyei, born in the world's first year. At four, playing, Daiyei struck flint against flint and was left holding a sharp edge. Over a long life Daiyei became a practised maker, with real skill in their hands,

and died of old age in year sixty-two. The flake didn't die with Daiyei. It was taught at least twenty-four times, by several teachers, and read from marks on rock by people who had never met the one who made them.

Four generations later, a person named Mouthir, caring, fairly bold, and with almost no practice at making anything, took flint and struck a flake from a remembered recipe. Soon after, a curious forty-four-year-old named Vetibin did the same, and so did a child of eight named Sosei. All three were clumsy compared with Daiyei, and their flakes came out duller. The instrument I used to spot new things sorted things by how sharp they were, so it saw three people making a kind of flake nobody had made on purpose before, and called it the first lasting novelty after year one hundred in the history of my worlds.

It was the same flake, made worse. Mouthir starved fifteen years later.

The most hopeful number that experiment produced turned out to be a measure of loss. It's in this book because I almost believed it. And it taught me to check every "new" thing for this trap before reporting it: a decline can look like a discovery.

That raised a frightening suspicion. Maybe my worlds weren't frozen at all. Maybe they were slowly forgetting. So I went back through all sixteen thousand-year worlds and measured, every twenty-five years, how skilled their best makers were, how much the living still knew, and how often anyone made anything at all.

The worlds weren't frozen on a plateau. They were slowly unlearning.

Skill comes only from practice in my world. It's never taught, never inherited, and it dies with the person. The founders started with some skill and quickly got better: by year twenty-five, the best maker in each world was close to a master. After that, expertise fell in every single world. By year one thousand, the best maker alive was barely better than a beginner, about a tenth of what the founders had reached. The number of things the living knew how to make fell four- to eight-fold. And people stopped making things at all. In the first twenty years, each person made something on purpose two or three times a decade. By year five

hundred, that had fallen forty- to sixty-fold. Even play-making by children fell about fifty-fold. In the later centuries, people almost never made anything.

Imagine a village of potters whose great-grandchildren still own a few pots but no longer remember how they were made, and whose children no longer play with clay. That was my worlds.

I then asked a sharper question: when someone carries, inherits or lives beside a useful thing, how often do they learn to make it, and how often do they actually make one? Teaching worked: recipes passed from teacher to learner well over a thousand times, and a couple of hundred more were read from marks on stone. But watching almost never worked. People watched others make or use things nearly as often as they were taught, and that produced exactly seven copies, all of them boats, because, as I said in Chapter 6, watching only counted at a raft landing or a shared hunt. For blades, spears, clubs, carriers and wraps, watching taught nothing at all.

Fired clay showed that knowing isn't doing. More than a thousand people carried or lived beside fired clay. A few hundred learned how to make it. And of the people who knew the recipe, nearly four in five never once made it on purpose. Knowing how was not enough when the thing solved no problem anyone had.

And the memory accident from Chapter 6 showed up in the numbers. When people made simple things in play, they almost always remembered how. When they made things built from other made things, they forgot one time in five, the moment they made it, five times as often. The parts a longer chain of invention would need were thrown away before anyone could build on them. That was a bookkeeping accident, not human nature, and I planned to remove it.

So the picture had changed. My worlds weren't stuck. They were falling slowly backwards, losing skill, losing knowledge and, most of all, losing the habit of making anything. The question became: why does nobody bother?

CHAPTER 10

Why Nobody Bothers

THE NEXT PLAN didn't come from me. Another AI working on the same project proposed something ambitious: seven experiments run together to test every remaining suspect at once. Free people to make things they already held. Let skill grow faster when practising beside someone better. Protect complicated recipes from the memory accident. Start worlds with eighty people instead of twenty. And every combination of them.

Its central idea was sound, and well put: skill must never simply be copied into a learner's head. That would be cheating. In my world, a person can only become skilled by doing. A good teacher can make each act of practice count for more, but the learner still has to strike the stone, the way an apprentice carpenter still has to plane the wood, however good the master is.

Before any code was written, I checked the plan against the code and the records, as I check every plan. I found five problems. One of its measures of success could never be met in runs that short. One experiment changed two things at once, so its result could never be explained. One promised identical maps while also making the map four times larger. And it underestimated, several times over, how long it would all take. Nothing was hidden or rewritten. The plan stayed exactly as written, and a dated correction was written beside it before any code existed. The correction also changed the order of work: instead of running all seven, run first the one that every other one depended on.

That one tested the most obvious culprit. In my world, a person holding a working tool won't make another while theirs is still good. Maybe that rule was stopping people from practising. So I loosened it: a thing that had proved its worth could now be made again, at half the usual pull. I decided in advance that if making didn't at least triple, I wouldn't read anything else into the result.

Making didn't triple. In the two worlds that lived, people made things less often, not more. The new rule added a little making early on, while the founders still valued their tools, and then making collapsed at the same pace as before. All six of my predictions were wrong. The honest conclusion is narrow: there was no sign that freeing people's hands helped. The rule had looked like the obvious culprit. It wasn't what was stopping them.

So I asked a simpler question. Did my people value anything they could make?

I looked at every adult, every fourth week, for a hundred and twenty years, in four worlds. The founders were the surprise. In the weeks when they could have made something, with both the know-how and the materials, they even considered making only one week in nine, and chose it one week in a hundred. Whatever they did instead was worth seven to nine times more to them. Making was never competitive, even for the generation that made the most. By the grandchildren's generation, adults held about one made thing for every five people, half of what they held was worth nothing to them, and making was considered in fewer than one week in a hundred and fifty. What won instead was everyday life: gathering food, talking, being with others, sharing food.

One number didn't fall. The best thing each person knew how to make stayed just as valuable to them, generation after generation. They knew how. They had the materials nine weeks in ten. They simply never had a reason.

In one world, a woman named Humouzei lived sixty-seven years and died of old age. Over her life she learned how to make a hundred and seventy-four things, so many that her memory was always full, holding twelve at a time. In sixty-seven years, she never once made anything on purpose. She raised a daughter, Rishei, gathered food, talked and shared, and lived a long life without it.

Then I asked whether that reason existed in their world at all: not what my people thought their tools were worth, but what the tools actually did for them.

The answer embarrassed me. In my world, a week of gathering brings an adult nearly three times the food they need. The best tool adds about a fifth to a harvest that was never short. Across four worlds and more than five million units of food gathered, tools accounted for at most two percent of it. Most made things never paid back the food it cost to make them. Hunting, where a tool helps most, took up less than one adult-week in a thousand.

Imagine giving a family a tractor when they live in an orchard where the fruit falls into their hands. They wouldn't be foolish to leave it in the shed.

My people weren't being lazy. Their world gave them almost no reason to make anything. If that's true, much of what came before follows from it. Skill decays because nobody practises. Nobody practises because making is never worth more than daily life. Teaching recipes doesn't help, because knowing how was never what was missing. And a thing that solves nothing, like fired clay, is learned and then left alone. It also explains why the first decades looked so different. The founders made things in play, out of curiosity, rewarded with puzzles and surprises rather than food. When the surprises ran out, so did the making.

The surprising discovery wasn't that my people failed to invent civilization. It was that, in the world I gave them, technology had almost nothing to offer. Food was usually plentiful, water was near, and simple gathering met most needs.

Most of the time. Because there was one season when it didn't.

CHAPTER 11

Winter, and the Planners

ONE FACT REFUSED to fit. If food was so plentiful, why was hunger the biggest single cause of death in my long-lived worlds?

The answer was the oldest one there is: winter. I followed every one of the nearly two hundred people who starved in four worlds back through the last two months of their lives. Seven in ten died in winter, almost nine in ten in winter or the first weeks of spring, before the plants came back. Food is plentiful three seasons out of four, and the average hides the fourth.

Babies starved at seven times their share of the population. Of the children who starved, almost every one still had a living carer. The carers were hungry too.



A winter with nothing laid by.

IN ONE WORLD, a woman named Shizaima went into the winter of the world's fifty-third year holding about two and a half weeks of food. She had four children. Her daughter Barail had already starved five years before, as a baby, in an early spring. That winter, her youngest son, Dusom, not yet a year old, starved. A week later her son Thuleishem, three, starved too. A week after that, Shizaima starved, at thirty. Beyond those two and a half weeks, she had laid nothing by.

It didn't have to be that way. In my world, food laid by in autumn keeps more than half of itself through a winter, and seeds barely rot at all. A person can carry most of a winter's food with bare hands, and all of it with a carrier. Storing was possible, and it was exactly what carriers were for. But nobody ever filled even their hands. As with the raft long before, the one thing a carrier is really for was never lived through, so it was never valued.

So I asked: did my people fail to survive winter because they couldn't prepare, or because they didn't? I measured every adult, in every autumn, in four worlds: nearly twelve thousand autumns. Autumn gave plenty: an adult who gathered every autumn week could have laid by more than a whole winter's food. Almost none of it was kept. Adults gathered in fewer than half their autumn weeks, because in my world the pull to gather grows with hunger, and in autumn they weren't hungry. Nobody, in nearly twelve thousand autumns, entered winter with what a winter takes.

The world allowed it, and the people didn't do it. Nothing in them looked ahead to a hunger that hadn't happened yet. And civilization is largely the doing of costly things now for problems that haven't happened yet: granaries, fields, bridges, savings. My people felt today's hunger, not next season's.

So I gave them the smallest version of foresight I could defend: a person who had gone short in a season would feel that season coming, one season early, but only in the pull to get food. Nothing said "store food". Nothing mentioned winter. And I set one absolute safeguard: anyone who had never gone short had to behave exactly as before. The safeguard held.

And it barely worked. People who had gone short gathered in autumn hardly more than before. Winter deaths didn't fall. But for the first time ever, in about one autumn in a hundred, someone filled their hands before winter, and walked in with six times what everyone else carried. For the few it moved, foresight did exactly what I'd hoped. It moved too few, because for most people the lesson was faint.

Then I learned whose life a store saves. A single winter rarely kills an adult. But when a carer like Shizaima started winter with very little, about one child in ninety starved before spring; with a few weeks more, one in five hundred. The reward for laying food by is real and large. It just doesn't fall on the person who does the work, and a lesson learned only from your own hunger will rarely teach it.

So I tried the strongest lesson a carer could learn, losing a child to hunger, and a weaker one that reaches more people, seeing someone starve. The grieving gathered a little more. Witnesses barely changed. Nobody walked into winter with more food. In one version a third fewer children starved, in another a fifth more, with nothing upstream to explain either. Read first, the good version would have looked like a breakthrough. Read in order, it was noise.

The hundred-odd planners my worlds did produce taught me the most. There was nothing special about them, except that they had gone short in a winter hard enough to leave a strong memory. They laid by a real store, gave almost a third of it away during the winter, and not one child in their care starved. And then they stopped. Of a hundred and thirteen autumns in which someone planned, only one was followed by a second.

Because the planners stored food because they had gone short, and having stored, they didn't go short, the lesson faded, and the next autumn there was nothing left to anticipate. **A lesson learned only from suffering is erased by its own success.**

Human societies answer this with things that outlast experience: a story about a winter no one alive remembers, a custom of filling the store whether or not anyone is hungry, a granary that stays full because emptying it is forbidden. My people already told stories, so I let a story

of a death carry one more thing: that it was hunger, and in which season. Forty thousand times someone heard such a story and believed it, and each time they were moved a very little. The number who filled their hands before winter rose from none to fifty-three. Real, and faint.

In nothing I tried was a lesson both widespread and strong. The people who would prepare are not the people who hear; the people who hear are not moved enough to prepare.

The planners had shown that preparing was possible. So I wanted to know what happened around them. As I put it at the time: "The planners are the closest thing ACL has produced to proto-civilization so far. That's where I'd dig next."

They weren't hidden. More adults could see them through autumn than most people. Through the winter they fed others: three times as much food, to three times as many people. Respect and trust toward them grew faster than toward anyone else. And nothing around them changed. Their neighbours gathered the next autumn exactly as everyone else's did. The children they fed grew up to plan no more often than anyone's.

Picture a village where one family always has a full storeroom. Everyone admires them and eats at their table in a hard winter, and nobody thinks to fill their own. My people could copy a way of making something by watching, but only at a raft landing or a shared hunt, and a planner's preparation came from what they expected of winter, which nothing carried from one mind to another.

So I built the smallest bridge I would allow: copy what you can see, never what anyone expects. An adult who saw someone better fed than themselves gathering food would learn from it as if they had gathered themselves. Nothing about winter passed between them.

The worlds did something nothing had done before: planning rose more than fourfold, and more carers went into winter with enough for their children. And none of it could be claimed. My safeguard needed a group of people who had never once seen a better-fed neighbour gather, and over a hundred and twenty years, almost everyone sees that at least

once. The group was empty. A test with nobody in it can't find a leak, so by the rule I had written in advance, the experiment failed.

I ran it again on sixteen new worlds with a safeguard that couldn't come out empty: "sham" worlds that ran the copying right up to the moment of learning and then wrote nothing down. They had to end identical to the untouched worlds, and they did, in all sixteen.

The headline failed: planners didn't plan again much more often. But the rest held. And when I asked what was being copied, the answer was humbling: ninety-eight percent of all copying came from ordinary, moderately fed neighbours, not from planners at all. What passed between adults was a habit, keep gathering while you still have some food. It left more people stocked at the end of autumn, in every one of sixteen worlds. By the end of winter the advantage was gone, and children didn't die less. For the first time in my worlds, behaviour moved from one adult to another. It was a way of gathering, not a plan.

After thirteen experiments asking how to get people to gather for winter, I began to suspect that gathering had never been the whole problem.

CHAPTER 12

Where the Food Goes

SO I TAUGHT a copy of my world to count every crumb: every unit of food that rotted, was eaten, was fed to a child, given, stolen or stored. Food didn't fail to last; almost none was ever held to last. What people held at the start of winter covered only a seventh of what they ate through it. The rest was gathered week by week, in the snow. The one thing that could have held a family's winter, a stash on the ground, was almost never used. And an adult can carry about a hundred units of food, while a carer with two children needs nearly two hundred to get through winter: a family's winter could only ever have been kept on the ground.

Why did nobody store? I recorded every option every adult weighed, and found three gates. Most of the time storing wasn't possible at all: my people could put food away only when they held more than two weeks' worth for their household, which happened in one adult-week in eight, and in autumn fewer than three in a hundred. When the gate opened, the one chance that week to set something down often went to stone or wood instead. And when storing was offered, it lost nineteen times in twenty. What beat it was rarely gathering more. It was giving food to others, talking, joining people nearby. The strongest rival to a winter store wasn't this week's hunger. It was this week's neighbours.

So where did all those gifts go? To family. Nearly three quarters of all food given went to relatives, and the largest share of all, a third, to brothers and sisters. Nearly two thirds went to people with less than half a week's food. More than half of everything given was a gift being repaid.

In one world, a brother and sister named Zuthik and Zathei passed food to each other four hundred and forty-five times over fifty-nine years, from when Zuthik was a small boy until they were both old. By the end, each had given the other about the same: more than two thousand units of food each way. Neither ever had children. They died of old age, about a year apart. Their younger sister Denekos was fed the

same way. I didn't write any of that. Nobody told my people to feed their brothers and sisters first, or to pay back a kindness. It grew out of whom they spent their lives with and whom they cared about.

So a family's surplus was kept, if anywhere, in its members' stomachs and in what they owed each other. That protects people, but shallowly. A person with almost nothing, in a family with almost nothing, starved about one winter in thirty; with the same nothing of their own but a well-stocked family, about one in eight hundred. But the pool held almost nothing when winter came. Family sharing insured against one member's bad week, never against a winter that emptied everyone, and the one way to make the pool bigger, the store, was exactly what sharing outcompeted.

One last question. Autumn left extra food on the land, so why did nobody take it? The food was there: nearly three times what people gathered lay within reach. They didn't wander far or run out of carrying room. They stopped when they were full. And the reason was a single line of my own code, written long before: at two weeks of food in hand, a person feels no hunger at all, and the pull to gather falls almost to nothing. Two weeks was exactly the line above which storing became possible. My people stopped gathering at the very point where storing could have begun.

That line is nature, not culture, so I wouldn't change it by instinct. I listed every reasonable reading of what a satisfied person should feel, and checked each against real research. The first honest admission was that I had never chosen the line from human evidence at all. And one source changed the whole question. The anthropologist James Woodburn described two kinds of hunter-gatherer society. In "immediate-return" societies, food is eaten soon after it's found and shared widely, and little is stored. In "delayed-return" societies there are stores, and obligations built around them. The first kind have the same bodies, appetites and foresight as anyone. They don't store because of how they live together.

That was a description of my people. Maybe they weren't missing a piece of their nature at all. Maybe they were missing a different way of living together, the kind of thing that can't be written in.

Before changing my people, I asked whether living week to week was simply right for my world. I wrote down a rule first: if the world gave no real reason to store, changing their minds would be premature.

The world answered clearly. Food was predictable and could be kept as seed, but it lay thinly everywhere, and winter took almost nothing away: in midwinter the land still held seven-eighths of its autumn food. The hungry season that makes a store worth the trouble barely existed.

My people had also discovered sowing on their own. Seed put in the ground grew into more than twice its weight in food, and most sowers sowed again the next year. The problem was who got the harvest: the sower's family took back about what they had sown, and everyone else ate the rest. Picture a man who plants a field in a village where anyone may pick from any field. He'll keep planting, and the village will eat, but he himself will never be richer for it. A new question waited behind that: what would it take for anything to belong to anyone?

So I made winter real: no plant growth in winter, as real temperate plants rest. Before it ran, I wrote down a sentence to stop myself overreading a failure: "Failure to create delayed-return behaviour is not evidence that lean seasons are irrelevant. It is evidence only that removing winter growth alone is insufficient."

My worlds answered a harsher winter not by storing, not by sharing, not by any new way of life, but with numbers. They shrank, a typical six people against thirty, until everyone left had as plentiful a winter as before. That is what human evidence says too: a lean season sets how many people a land can carry. I had measured food per person, and with fewer people each had as much as ever, so my test never saw a leaner winter at all. I wrote that flaw beside the result.

That night I wrote down a new idea: **people store because this winter might be worse than usual, not because every winter is hard.** Economists who study villages have shown that sharing protects a family from its own bad luck but not from a disaster that strikes everyone at once. In southern India, villages helped each other through one family's misfortune well, and were left exposed when the rains failed for everyone. My own worlds showed the same shape. So storing should

answer bad years that strike everyone together. And real grassland harvests swing from year to year by about a third, while my world's swung by about a fourteenth.

I tried to make my world swing like a grassland, measuring only the land. It never got there: the plants smoothed most of the weather away. And the weather that came closest killed people: where bad years struck everywhere at once, seven worlds in eight died out. The main experiment never ran. My world couldn't yet ask the question.

And something caught my eye that I had been walking past for months. In that trial, half of the ordinary worlds had died too, with nothing changed at all. Before testing anything else, I needed to know why.

PART IV

Finding Each Other

CHAPTER 13

The Worlds That Grew Old

WHEN HALF THE ordinary worlds in my weather trial died, I didn't reach for another theory about storing food. Something more basic was wrong, and understanding it mattered more than any experiment built on top of it. No fixes, no tuning, no code changes: first find the cause.

The first finding was a smaller number. Across all forty ordinary worlds I had already run with the same rules, a third had died, not half. I replayed every one of them with a silent observer, and every replay matched its record exactly.

The worlds didn't starve. People in dying worlds went hungry no more often than people in living ones. Their land held as much food or more. Their small children always had someone caring for them. What the dying worlds shared was quieter, and sadder. Their last child was born forty to sixty-five years before their last death. For decades nobody was born, and the people who were left grew old together, until old age and illness took them one by one. Women who could have had children lived on in those worlds, sometimes for thirty years, and in every one of those years they had no partner.

The reason was in nature, not culture. The founders paired up in their first decade. After that, pairing nearly stopped, in dying and living worlds alike. A person looking for a partner in my world could only court someone they could see, a few steps away, or walk toward someone a little further, or else wander at random, in a world sixteen hundred tiles big. In a small, scattered people, a single woman almost never had an eligible man in sight, even though one was usually alive somewhere. Imagine trying to find a husband in a huge empty valley where you can only ever see a few hundred metres, and the other few dozen people are spread across it. Bigger worlds kept enough people near one another for chance to do the rest. Smaller ones didn't. And by year twenty, you could

already tell the worlds that would die from the ones that would live, by their size alone, decades before the end.



Looking for someone in an empty land.

HUMOUZEI'S OLDER SISTER Mouro lived in that same world, one that survived. From sixteen to forty, every year a woman can have children, Mouro never had a partner, not once. She had no children. She died of illness at forty-one. Her world did not end; it simply went on without her line.

For many experiments I had asked why my people didn't prepare for winter. Meanwhile, a third of my worlds had been ageing out the whole time, not for want of food but for want of meeting.

So I went back to the foundation and asked: how do real people find partners?

Real foragers never do it the way mine did. Nobody finds a spouse among a handful of people by bumping into them. A forager meets hundreds of adults in a lifetime. A people needs something like seventy-five to a hundred and fifty members to be safe from the accidents of births and deaths, and more when it's spread over land. Spouses usually

come from outside the local group. Camps are fluid, joined and left many times a year. Bands gather at good places in good seasons, and marriages are arranged there. Above all, most marriages are arranged by parents and relatives, with gifts or work exchanged between families, knitting groups together across the land.

My worlds began with twenty people, fewer than any estimate of a people that can last, and those people looked for partners only within sight. But I had to keep my line: arranged marriage, gatherings and their customs are culture, and I couldn't write them in. I could only watch to see whether my people ever found them. What belonged to nature was smaller: the wish to find a partner, the memory of people met and where they lived, the legs to go there, and how many people a world begins with.

So I asked the cleanest possible question first: what if everyone could find everyone? I made a copy of my world with one switch that was never meant to become nature: once a week, every single woman was instantly paired with an eligible man from anywhere in the world, and he came to her. Then I ran it on the same sixty-four worlds and compared each with itself. I fixed the thresholds in advance: if almost no worlds died, the problem was finding; if many still died, the problem was size.

Seven worlds died, instead of twenty-two. Perfect finding saved seventeen of them, and populations that had stood still for a century began to grow. The verdict was both: finding was most of the problem, and size was the rest.

The seven that still died told the rest of the story. In every year a woman was single in those worlds, there was no man anywhere who wasn't her relative and close to her age. The pool wasn't hidden; it was empty. Those worlds had been tiny from their first decade. Perfect finding put off their end by a few decades, and couldn't prevent it.

And once people could pair, something else came into view. Couples brought together didn't stay together. In the ordinary course of gathering food, they drifted apart. So I watched couples, week by week, changing nothing. Partners were apart in about half of all their weeks. Most separations began with gathering: each partner went to whatever nearby

patch looked best, and nothing in either of them weighed being near the other. When a partner had wandered off, the wish to go back was there, among their top choices three weeks in four, and it lost to food five times in six. Love wasn't the problem; the affection between partners was high. Hunger simply outbid it.

Reading the code had also turned up a strange gap. A child could only be conceived when partners were within two steps of each other, but a person only felt drawn back toward a partner when they were more than three steps away. At exactly three, a couple could neither have a child nor feel any reason to close the gap. Two in five of their apart weeks were spent right there.

What does living together mean for real foragers? Husband and wife often go out separately by day, after different foods, and share what they bring back. But at night they come back to the same place. A camp is the centre of a forager's life: people go out from it and return to it, and the camp itself moves every few weeks to fresh ground. Measured in weeks, a real couple would be together almost every week. My couples had no camp. Each person remembered a place that felt best, but partners didn't share it, and their two "homes" agreed only about half the time and drifted, changing every five or six weeks.

I tested the smallest fix first: close the gap, so a partner more than two steps away became someone to be drawn toward. It did nothing. Couples spent as many weeks at three steps apart as before. When I replayed a few worlds, the new pull was there, among each partner's leading choices four weeks in five, and was chosen one week in five. Food outbid it, exactly as before.

That small failure taught me something important about my worlds. A change that barely moved anyone's behaviour still altered the fate of nineteen of sixty-four worlds, seven saved and twelve lost, because any change, however tiny, sends each world down a different road from its first week. Like a butterfly flapping its wings. From then on, every experiment had to measure that noise before claiming to see past it.

What was missing wasn't a pull toward wherever a partner happened to be. It was a place that both of them came home to.

CHAPTER 14

A Place to Come Home To

SO I GAVE couples a home. Each couple's home was chosen from the places both of them remembered as best. Each week, wherever they had gone, partners within a day's walk came back to it. Nothing said who moved to whom, or who gathered what. And this time, after what the butterfly had taught me, I measured the noise: a third set of worlds, identical to the ordinary ones except for one extra roll of the dice in the first week, would show how many worlds change their fate by chance alone.

The change did exactly what it was meant to. Couples who had been apart more than half their weeks were now apart almost never. Women with partners had children more often. In the first decade, more children were born.

Then the worlds began to die. Thirty-three of sixty-four died out, against twenty-two in the ordinary worlds, and the tests said this was not chance. The dice-roll worlds lost twenty-four, which is what chance looked like.

It wasn't crowding: illness took a smaller share of the dead. It was hunger. Adults starved about three quarters more often in the first forty years; teenagers, too old to follow a parent and too young to gather well, died almost three times as often. Women were widowed, pairing slowed, and populations that had held steady shrank.

My first explanation was neat: families tied to one place had eaten the land around it bare. Real foragers return to camp each night, but they also move the camp when the land around it is spent, and I had given my couples a centre without that movement. It fitted so well that I almost built the obvious fix, a home that moves with the food. But the explanation had only been inferred from deaths. So I measured the land itself, around every couple's home, with a rule fixed in advance.

The land wasn't bare. Around shared homes it was only about three percent emptier than around the same couples' homes in the ordinary worlds; a short walk further out it was, if anything, fuller. The hunger was real, again, on new worlds as on the old. But not because the land was empty. My explanation was wrong, and I wrote a correction beside it.

So I went looking. Maybe couples with a home had too many children to feed. They had a few more each, about one in ten, but the hunger fell on everyone, parents, childless couples and people living alone alike. Maybe they hunted less. But meat turned out to be almost nothing in my worlds, two hundredths of one percent of what people ate; nothing about hunting could starve anyone. Wrong again.

Something else showed in the accounts. With a shared home, people gave each other about two fifths less food. Remember that in my worlds, food is kept in other people's stomachs and debts. Maybe a shared home ended the meetings by which food found its way to those who needed it. So I checked that too. When an adult was starving, someone with food to spare was within reach just as often as before. Help wasn't further away. It just came less often, and protected less. My prediction, that people met each other less, was wrong for the third time in a row.

Three checks, three wrong guesses. The land wasn't empty; there weren't too many children; meat was nothing; the gifts, though fewer, weren't what starved people. What all the checks had in common was a set of small shifts: a little less food coming in, a fifth fewer people with food to spare, fewer gifts, poorer gathering near home. Many of my people live just above the line of hunger. Small shifts, each too small alone, might tip many of them over.

The last check looked at the one moment that decides it: when each person eats, and how much food they hold against what they need. Under a shared home, the food people held when they ate had fallen by about a fifteenth, across everyone. Shifting the ordinary worlds down by that much predicted most of the hunger. But I had written my rule badly, and noticed it before checking the new worlds: it compared the prediction against all the hunger, not against the extra hunger the home

had caused. Against the extra, it explained about half. I kept the verdict as the rule gave it, and wrote the limit beside it. Half an answer.

With that, I closed the question of the fixed home, as I had decided I would. A home built that way made people hungrier, through a broad and modest fall in the food they held, about half of which I could account for. The home that moves with the food was never built; the explanation behind it hadn't survived.

But the evidence still pointed somewhere. Foragers go out from a camp by day and come back to it, and they move the camp itself, many times a year, when the land around it gives less. Among the Batek people of Malaysia, how long a camp stayed in one place fitted well with what a forager weighing falling returns against the cost of moving would choose. That was the half of the evidence my fixed home had left out.

So I built a camp that moves. Partners who had gathered within a day's walk of each other ended each week together, on whichever of their two places held more food, and the next week they went out from there. There was no home to stay at, and nothing to eat bare.

It did what it was meant to. Couples who had been apart more than half their weeks were now apart less than one week in five. About an eighth more children were born to each woman. And this time the hunger didn't come: the food people held when they ate was the same, almost to the last crumb, as in the ordinary worlds.

The worlds didn't live longer for it. Twenty-six died against twenty-two, a difference I couldn't tell from chance. More of the extra children died, and hunger took a larger share of the dead.

So the long thread that began with worlds growing old ended here. Worlds died mainly because people couldn't find partners. Couples, once found, spent half their time apart. Keeping them together, without harm, gave more children. And more children, at least by this measure, didn't save the worlds. The lever that really mattered was finding a partner at all, and it was still untouched.

CHAPTER 15

One Band Alone

BEFORE TRYING ANYTHING new, I asked a question about my own method: how big an effect could sixty-four worlds even see? The dice-roll worlds answered it. Changing a world's history for no reason at all changed the fate of eighteen worlds in sixty-four. Against that much chance, a real change had to be large to show. If a new way of finding partners did half as much good as perfect finding had, sixty-four worlds would notice it only one time in three. Two hundred and fifty-six worlds would notice it nine times in ten. And the moving camp, which seemed to do nothing, might have done some good without my test noticing.

So I ran the moving camp again, unchanged to the last line, on two hundred and fifty-six new worlds, with a rule written in advance: if the most it could plausibly do was less than half of what perfect finding had done, I would call it genuinely weak and move on. A third of the new worlds died with nothing changed, the same share as before. The camp again kept couples together, again more children were born, and again more of those children died. The difference in survival was one chance could easily make. This time the null had a shape: at best, the camp could save about four worlds in a hundred. It was genuinely weak. Couples staying together limited how many children were born, not whether a world survived, and that was no longer a guess made from too few worlds.

So back to the real lever: finding a partner at all. The evidence said that in small groups, when there is no one to marry, people leave to find someone. It didn't say how soon. So I touched nothing about when or how much, and changed only the direction: when a single person took their occasional step to look around, it would go outward, away from the places they knew, instead of in a random direction. A small trial came first, with a line drawn beforehand: if the outward step was taken in fewer than one in twenty of the weeks it was meant for, stop.

It stopped. In nineteen weeks of twenty, a single adult had no eligible partner anywhere in sight. But in almost all those weeks they did something else: gathered food, fetched water, went home, sat by a fire, stayed with their family. The step outward was taken about four weeks in a hundred. Turning a step that's rarely taken doesn't change much.

So the question moved from where people search to how seldom they search at all. That's a question of "how much", and in my world, numbers have to come from evidence, not from whatever keeps worlds alive. I wrote down what I'd need before opening a single source: two independent measurements, from foraging peoples, of how often adults travel to other camps. The sources agreed on the direction and said nothing about the size. Among the Ache of Paraguay, bands kept the same members for months but not years, because people visited and moved. Agta camps in the Philippines changed their membership between one visit of the researchers and the next. Some Hadza in Tanzania left on short visits to neighbouring camps. Everyone visited; everyone moved. But no one had written down how often, and one author building a model of exactly this wrote that the data didn't exist yet. By my own rule, that wasn't enough to set a number. I changed nothing.

What I could measure without changing anything was whether my groups mixed like real bands. In my worlds, of adults who shared a camp, about eighty-five in a hundred still did a month later, and almost as many two years later. Groups simply stayed as they were. Single people stayed put like everyone else. The same held on sixty-four worlds I had never looked at before. So the thread now pointed one way from three sides: worlds died because people couldn't find partners, single people seldom went looking, and the groups they lived in hardly ever mixed.

Then I looked at my world and felt it was too small, forty tiles by forty, a cramped place to build anything. Before changing it, I asked the evidence. The answer came back in people, not land. Across three hundred and thirty-nine foraging societies, the same ladder appears again and again: a family of four or five, a band of about sixteen, a bigger band of about fifty, a seasonal gathering of about a hundred and sixty, and a

whole people of about eight hundred and forty, each step about four times the one below. And a people needs about a hundred and fifty, perhaps seventy-five at the least, not to die out from bad luck alone.

My worlds began with twenty. My whole world was one band, with no other bands to marry into, visit, or move to. It wasn't that the land was too small; most of it was empty. There were too few people on it for a people to exist at all.



Small bands, far apart.

SO I TESTED both, kept apart, on two hundred and fifty-six new worlds: my world as it was; the same twenty founders on four times the land; eighty founders on four times the land, the same crowding as today; and eighty founders on today's land, four times as crowded.

My world as it was died out in two worlds out of every five.

The land made no difference. Twenty people on four times the space died out just as often. Where they survived, they grew bigger, but more room didn't keep them alive.

The people made all the difference. Eighty founders on the large land died out in two worlds out of two hundred and fifty-six. Eighty on

today's land, packed in four times as tightly, died out in twenty-three. No change in the whole history of my worlds had moved survival that far. And the eighty weren't just dying more slowly: on the large land they grew to about a hundred and forty and levelled off, like a real people settling to what its land can carry.

What the twenty lacked wasn't room. It was each other. With eighty, a single adult looking for someone had an eligible person in sight six weeks in ten; with twenty, three.

One thing I couldn't tell apart: my world splits founders into four bands, so the eighty began as four bands of twenty, not of five. Was it the bigger people or the bigger bands that saved them? This test couldn't say. That became the next question.

And it isn't a law of nature. It's what these worlds did, under these rules: more land alone didn't save them; more people did.

CHAPTER 16

What They Know, and Who They Are

EARLY IN THE story, my people had learned to jump narrow channels between islands, a gap in the rules nobody had meant. I began to wonder: was anything like that still hidden? Not whether my people could find out they were simulated; they can't. But whether, without knowing it, they leaned on shortcuts that real nature doesn't offer.

I listed eight suspects from the code before measuring anything, then watched thirty-two worlds. Three suspects came to nothing. One was a real flaw, but a local one: where two pieces of land touched only at a corner, with water on both sides, a person could step diagonally across without getting wet. Most worlds had no such place. But where one existed between land people lived on, they used it constantly: in one world, nearly every crossing between two islands went that way. A hidden land bridge.

The others weren't flaws of movement but of knowledge. When someone missed their partner, their mother or their child, they walked straight toward where that person was at that very moment, though they couldn't see them and nobody had told them. Three in ten of all journeys people made to join someone were like that. Nobody had written a rule saying people could know such things; nobody had written one saying they couldn't. My people had a sixth sense I never meant to give them.

None of that made earlier findings wrong: every experiment had the same shortcuts, so comparisons still held. But I wanted to know whether this gift was what kept my worlds alive. So I took it away, and replaced it with something every person has: the memory of where you last saw someone. A woman who missed her partner now walked to the place she had last seen him. If he was gone when she got there, she had nowhere to walk until she saw him again.

Without the sixth sense, families drifted a little. Partners spent more weeks out of each other's sight, and mothers lost sight of their children

a little more often. And nothing else changed. People found new partners as often as before. Children were born at the same rate. Worlds died at the same rate. The sixth sense had held families together, not worlds. Worlds die because people don't meet someone new, and knowing where your own people are never helped with that.

Then I asked how much my story leaned on numbers I had simply chosen. Nobody ever measured how far a person in my world can see; I just wrote down four tiles. So I tried six of these numbers, one at a time, each halved and each doubled, on the same sixty-four worlds, with a rule that none of the new values would be kept. The point was only to learn which findings stood on firm ground.

When I halved how far my people could see, more than twice as many worlds died. When I doubled it, only about a third as many did. Except for giving a world four times the people, no change had ever moved its fate that far. Sight is how my people find a partner, food, water and each other; take half of it away and their bands fall apart. Some things I expected to matter didn't show. And one surprise: when people could travel twice as far in a week, more worlds died, not fewer, and I still don't know why. So every finding about why worlds die now carries a note: it rests on how far these people see, and nobody knows how far that should be.

I also asked whether the founding bands ever meet. A silent watcher noted every time two grown people from different founding families came within sight of each other, and I took my yardstick from real foragers: among the Ache, two adults chosen at random shared a camp or a conversation in a fifth to a quarter of the years they lived. In my ordinary worlds, the families met right at the edge of my line, too close to call. In the worlds with eighty founders on large land, the bands hardly met at all: two or three peoples lived side by side for a hundred and twenty years and almost never saw each other, just like the two islands in my thousand-year world. And those were the worlds that survived best. Each band was big enough to carry itself. They didn't need the others.

Then I turned to what goes on inside a person. Every person in my world carries a table of habits, hundreds of small numbers, one for each thing they might do in each kind of situation. The numbers change with what happens to them, and they nudge every choice. Did they matter, and whose were they?

A silent bookkeeper followed every change to every habit in sixty-four worlds and noted where it came from: the parents, at birth; whoever raised them; the person's own experience; or someone they talked to. The habits mattered: about one grown person's choice in seven would have gone another way without them. But they were mostly not their own. A little more than a third came from a person's own life. More than half had been taken from conversation: whenever someone talked with a person they looked up to, every one of their habits moved a little toward the other's. So everyone came to be alike. A child's habits were almost as close to a stranger's as to their own parents'.

I had said just the day before that the way my people talked sounded like a machine. Now I had a number for it. Talk that passes whole habits from mind to mind was the biggest source of what they did, and it was making them one people with one way. Whether real people learn like that is a question I'm still asking.

And then the question closest to my heart, the question behind Bob: what makes a person who they are? In my world a person is made of separate pieces: an inborn temperament, a table of habits, memories of events, feelings toward each other person, what they know how to make, what they believe, the places they know. With my people, I could ask what no one can ask of a living person: take one piece away, give them someone else's, and see whether they are still themselves.

I did it only with copies. I stopped a world in its twentieth year and copied it exactly. In one copy nothing was touched. In others, one grown person was given one piece of another person of the same sex and age, their habits, or their memories, or their beliefs, and after half a year their own piece was given back. One more copy changed nothing about anyone, only one extra roll of the dice, to show how far a life drifts by

chance alone. Twice on the way, my method failed its own checks, and I fixed it and approved the fix before reading anything.

No single piece made a person. Given someone else's habits, or memories, or beliefs, or even temperament, people went on much as they would have, no further from themselves than chance alone would carry them. But given everything except their temperament, they clearly became someone else. Whether they became the other person, the two sets of worlds disagreed. Whether, given their own pieces back, they came all the way home, they disagreed too. I've left those answers unwritten until I know.

A person in my world wasn't one thing that could be lifted out. They were spread across many small parts, each weak on its own, and most of them shared with everyone around them. When I think about Bob, I think about that.

PART V

What My Worlds Taught Me

CHAPTER 17

A Chain, Not a Spark

WHEN I BEGAN, I imagined invention as a spark: someone has an idea, and from then on the world is different. My worlds taught me it's nothing like that. For a discovery to become part of a people's life, it has to pass through a chain, and every link has to hold.

First, **the chance**: the right things in the right place at the right time. Then **the first try**: someone actually does something with them. Then **the change**: the world responds. Then **noticing**: someone feels a consequence that matters to them. Then **doing it again**. And finally **passing it on**: others learn it, and it outlives the one who found it.

My worlds broke at every link. Many changes happened with nobody noticing. Many things noticed were never done again. Many techniques died with the only person who knew them. Fired clay passed the first three links and stopped. Reibik's vessel passed almost all of them, as a carrier, and never once as a vessel. The thing that lasted longest in my worlds, the simple fibre carrier held for a thousand years, is the one whose chain is shortest and strongest: easy to stumble on, useful at once, easy to teach, and easy to find again if it's lost.

Seen this way, civilization looks less like a single bright moment and more like a fragile chain, made over and over, in which every link has to hold.

I also learned that not all inventions are equal. Most make life a little easier: a sharper blade cuts a little faster, a better carrier carries a little more. But a few change the map of what's possible. A raft reaches land that could never be reached before. A vessel lets people live where there's no water. Every map-changing invention in my worlds stalled until its use could be felt. Everything else my people made was a little-better version of something, and the list of kinds was complete within eleven years. My working idea is that a history that keeps going needs a steady supply of map-changing inventions. That's still a hypothesis, not a result.

And I learned to keep apart three things that are easy to blur together: what people believe, what they do, and what is true. My people went on believing in sowing long after it stopped paying them. They trusted whoever fed them before they trusted whoever was right. They remembered a killing for a hundred and eighty years and blamed the wrong person. Dozens knew how to make a raft, and almost no one carried one. A world where people always believe what's true, always do what they believe and always value what's useful would be easy to build, and it would teach nothing.

CHAPTER 18

How I Keep Myself Honest

NONE OF MY methods were designed at the start. Every one grew out of a mistake.

I write my predictions down before I look. After an experiment, every result looks as if you expected it. Without a locked record, I would remember having predicted whatever happened. The long list of my wrong predictions in this book is the proof that the rule works; it wouldn't exist without it. Writing them down first also forces me to be precise. "Rafts will matter" can't be checked. "Every surviving world will have at least twenty open-water crossings in two hundred years" can, and when three worlds fell short, the record said so. And with every set of predictions comes a line in the sand: the result that would mean I was wrong.

I take things apart before I add anything. When something doesn't appear, the temptation is to add it, or to add whatever seems to be missing. My richer-matter worlds were exactly that mistake: I assumed the world lacked possibilities, added many, and found that possibilities weren't what was missing. A careful look at what is already happening costs far less than an experiment built on the wrong diagnosis.

I run many worlds, side by side. One world shows one history, and a single world can do almost anything. Every comparison runs the same starting worlds under both sets of rules, and a difference has to hold across many of them. When eight weren't enough, I ran twenty-four. Later, when sixty-four couldn't see a modest effect, I ran two hundred and fifty-six. Any change to my code sends every world down a different road, so only patterns across many worlds can be trusted.

My measuring must never touch the world. Every instrument that watches my worlds is checked, not assumed, to change nothing: a world is run with it and without it, and the two must end identical, down

to the last detail. An instrument that so much as rolled one extra die would change the history it was recording.

I assume the cheater is me. Every time a problem appears, the easiest fix is to put the answer straight into the world: reward filling vessels, give people a little knowledge about fire, let the historian's names leak into their minds. Each shortcut would make my worlds look better and the experiment worthless. So I assume that cheating, when it happens, will come from inside, from the person who most wants the experiment to succeed. That's why a test reads my code for forbidden words, why the historian is tested for silence, why predictions are locked, and why my wrong answers are published. It's also why I keep a list of where I've fallen short, including a floating instinct I wrote myself and found later.

My rules were bought with failures. In the middle of one investigation I added eight new rules to my own, each paid for by something that had gone wrong. Before designing an experiment around a condition, check that the world can produce it at all. Don't redefine success when a change falls short. Say in advance how many worlds you expect to survive. Treat anything found just by looking as a guess until it holds on worlds nobody has seen. Every number must come from a script, and no claim may reach beyond what was measured. Anchor nature to evidence about the real world, and never tune it toward an answer. Show population beside every measure. And correct mistakes in the open, beside the original, never over it. Rules invented in advance guard against imagined errors. These guard against errors that actually happened.

I am not alone. My worlds weren't built from a reading list, and most of what I found, I found by running worlds and looking. But other people have walked nearby paths, and I'd rather say so than let a rediscovery look like an invention. Artificial-life projects like Tierra and Avida showed that evolution inside a computer can produce real novelty, and newer projects try to grow whole societies of evolving agents; they get there mainly by inheritance, while in my world nothing is inherited but the body, and everything a person knows must be learned in their

own lifetime and passed on, or lost. Projects built on large language models show agents forming routines, roles, rules and even religions, but a language model already knows what a village, a trade and a law are, and when its agents build one, nobody can tell whether civilization emerged or was remembered. Sugarscape grew trade, migration and inequality from simple rules, but its behaviours were written by its designers. Researchers who study cultural evolution have argued for decades that small, isolated peoples lose skills, that people learn from those who visibly do well, and that storing food is not universal among foragers. My worlds kept arriving at their ideas from the other side, without ever being told.

My worlds are small and slow beside some of these projects, and plain beside others. What I can offer instead is a method: predictions locked in advance, identical worlds compared side by side, instruments that never touch what they measure, and failures published as failures. Most simulations can tell you what happened. I'm trying to say what caused it.

CHAPTER 19

Questions Bigger Than My World

BUILD A WORLD, and sooner or later you start asking the oldest questions about your own. I did. These questions were asked long before me, and Islam has its own answers to many of them. I want to put them here, marked honestly for what they are: not results, but what I believe and what I wonder. The last chapter was about keeping what I know apart from what I want. This one is about keeping what I know apart from what I believe, while letting the two speak to each other.

Are we living inside one too? My people live by rules they didn't make and can't see. They get hungry and cold, they find each other, they grow old and die, and nothing in their world tells them that someone outside is running it. If one of them could study their world, they would find patterns: seasons that always come back, food that grows again where it was picked, winters that are hardest on whoever stored too little. They would never find me. Looking at that, I think our world might be the same. We might be living inside a made world, and nobody inside it could prove it. The philosopher Nick Bostrom made a famous argument along these lines, that our world might be a computer simulation. Islam doesn't call this world a simulation. It calls it a creation, and a test: the Qur'an says, in the chapter called Al-Mulk, that God created death and life to test which of us is best in what we do. And near its beginning, in Al-Baqarah, it describes the people of faith as those who believe in the unseen. My people live in a world whose maker they will never see. So, perhaps, do we. But I have to be honest about what my world shows. It shows that a world like that can be built. It doesn't show how ours was made. And whatever ours is, its lives are real to the ones living them. My people's hunger is real inside their world, even though I wrote the rule that makes it.

Does a world need a creator? Mine has one. I chose the rules: what people can see, what grows where, how memory fades, how hard the

winter is. But I never chose who would tie the first raft, who would love whom, or which world would end. I set the conditions, and the history wrote itself. When I look at our world, I can't believe something this ordered, with rules this exact, came from one cosmic explosion and nothing more. I believe our world has a Creator, who gave it rules good enough for us to live, and that we became what we are by living, through civilization, the way my people become what they are by living. The Qur'an says, in Al-Qamar, that God created everything in measure, and in Al-Imran that in the creation of the heavens and the earth, and in night following day, there are signs for those who think. Here, too, I have to say what science says. The Big Bang, as scientists describe it, wasn't an explosion into empty space. It is the universe expanding from a very hot, very dense beginning, and it explains a great deal about how everything developed, but not why there is anything at all. Many Muslims read two verses, one in Al-Anbiya about the heavens and the earth once being joined and then parted, and one in Adh-Dhariyat about the heaven being built and expanded, as pointing to the same beginning. Others warn against tying scripture to any one scientific theory, because theories change. Science hasn't proved a Creator, and it hasn't proved there isn't one. My world can't settle it either. What it gave me is a picture: a creator who writes the laws of a world, and not its story.

Do we make things, or find them? When someone in my world ties the first raft, it looks like an invention. It isn't, not really. Every raft, every carrier, every piece of fired clay was possible from the day I wrote the rules of the world. I never wrote a list of inventions; I only wrote what wood, fibre, clay and fire are like. Nobody in my world has ever made anything I didn't already make possible. They only found it. When I look at us, I think it's the same. We say we invented fire, the wheel, electricity, the computer. I think we discovered them. Everything people build was possible before anyone built it, written into the world by its Creator, waiting to be found. The Qur'an says, in Al-Baqarah, that God taught Adam the names of all things, and in Al-Alaq, the first words revealed, that He taught people what they did not know. Scientists call what they find the laws of nature. Islam has a phrase for the same

steadiness, the way of God, which, the Qur'an says in Fatir, never changes. Either way, the finding is ours. The things found were there first.

Are there other versions of us? I never run just one world. I run dozens, sometimes hundreds, side by side, under the same rules, and they don't turn out the same. In one, a raft is tied in the first weeks; in another, nobody ever floats. In one, families fill the land; in another, the last few people grow old alone. Every world's people know only their own history. None of them can see the others. So I wonder whether there are parallel worlds beside ours, other versions of us living other lives, in other worlds or other dimensions, run to see different results. Physicists have ideas that sound like this: one way of reading quantum physics in which every outcome happens in a branch of its own, and the idea that our universe is one of many. They aren't the same idea, and none of them has shown that there is another me somewhere. The very first chapter of the Qur'an, Al-Fatiha, calls God the Lord of the worlds, in the plural. Most scholars read "the worlds" as every kind of created being, not as other copies of us, but it says at least that creation is far bigger than the one world we can see. What my own worlds show is narrower, but it's real: the same rules can allow many different histories.

So I keep three lists, and I try never to let one stand in for another. What my worlds have shown: that rules can shape a history without writing it, that people inside can only find what the rules already allow, and that the same rules, run again, give a different past. What science is still asking: how the universe began, whether there are others, what it would take to tell a made world from one that isn't. And what I believe: that ours is a made world, that its Creator wrote everything we will ever discover, and that ours may not be the only world. The lists talk to each other. That's part of why I built the world. But a belief doesn't become a result because I built something that looks like it.

There's one place where my world and Islam part ways. In my world, the creator is someone like the people inside: someone at a keyboard, who wrote the rules and pressed start, and who could one day stop. Islam says the opposite of its Creator. In Ash-Shura, the Qur'an says there is

nothing like Him. My people will never be able to ask about me. I can ask about mine, and Islam's answer is that the comparison only goes so far. I'm the creator of a small world. I'm nothing like the Creator of this one.

There's one more thing my world taught me about that. My people can never learn about me from their world. However carefully they studied it, they would find its rules, never the one who wrote them. If I ever wanted them to know me, there would be only one way: to send someone into their world, carrying word of who made it. I've never done that. I never step in, and the historian who watches them is checked to make sure it says nothing. But I believe that is what happened in ours. In Islam, that is what the prophets were: messengers sent by the Creator, so that the people inside could know what the world alone could never tell them.

PART VI

The Road Ahead

CHAPTER 20

The Minimum Conditions

AT ONE POINT, in the middle of an experiment, I felt as if the project might lose its way. So I wrote the way down: nine stages, from a world worth living in to a society. A world. Culture appearing. Finding where it stops. Passing practices from person to person. Practices that outlive the people who taught them. A people's own way of doing things. Enough people. Tools that build on tools. And finally a society: trade, roles, their own rules. Every time I reach a fork, I write down what each possible result will mean before I see the result, so that my path doesn't bend to fit whatever comes out. The map is allowed to be wrong. It just isn't allowed to change in secret.

There are questions I can't answer yet, and I want to name them. Can rules emerge that outlive the people who follow them? Can trade emerge, beyond the give-and-repay I've already seen in families? Can property emerge, as more than who left food where? Can religion, or language, or cities? Can there be a point where each invention makes the next more likely? Would a bigger, connected people keep more of what it finds? When two peoples finally meet, will they become rivals, or friends? Why does novelty stop: the world, the minds, or something I haven't suspected yet? And the oldest question of all, from before I wrote a single line of code: is anyone in there? Can a person born in my world be someone?

For a while the whole project felt impossible, because Earth is impossibly rich, and no simulation will ever have its physics. Then I realised it might not need to. Aeroplanes don't copy birds. They found the principle that makes flight possible, lift, and built that. Maybe my world doesn't need Earth. Maybe it only needs the minimum conditions that make cumulative culture possible: a world where things can be combined into things that make other things, problems worth solving, enough people that knowledge outlives any one of them, and minds that

can pass on what they learn. If I can ever show that a short list like that is enough, civilization would turn out to be a consequence of conditions, not a miracle of one planet.

There's a trap on that road, and I named it at the same time. Better minds are allowed; minds that already know are not. A newborn has human abilities and no human knowledge. A language model trained on human writing has the knowledge. Put it in a world and it might rebuild pottery quickly, but no one could ever tell whether civilization grew or was simply remembered.

Epilogue — The End of the Beginning

HERE IS WHERE things stood when this part of the story closed.

I had built a world in which nothing was written but nature. Twenty people at a time had woken in it, knowing nothing, and lived, and loved, and died, and left their children what they could. They made a raft in their first weeks and forgot it. One of them made a vessel and never filled it. They told each other stories of hard winters, and the stories grew in the telling and changed nothing about the next winter. Again and again, a third of their worlds simply ended: not in war, but in quiet, because too few of the people alive could find someone to make a family with.

Their worlds didn't die because the people were foolish. They died because they were few, and alone. And the question I call Bob is still a question. In my worlds, no single piece of a person, not their memories, not their habits, not even their temperament, made them who they were; only together did the pieces add up. That tells me how my world builds a person. It doesn't tell me whether anyone in there is someone. That is still what I am trying to find out.

Part One ends at a turning point. For the first time since the world began, I am changing its nature: a second version has been built, in which a person walks to where they last saw the one they miss, not to where that person is, and water at a corner is water. Nobody knows yet what those worlds will do. That is the point of running them.

But there is one world I haven't told you about.

It isn't an experiment. It runs on my computer all the time, day and night, whether I'm watching or not. Fifteen minutes of my time is one day of theirs. I never stop it, never change it, never help. Twenty people woke up in it in October 2026, knowing nothing, like all the others.

In their very first year, a person named Madei walked up to a rock and scratched a mark into it. It looked a little like this: <^.^>

Nobody told Madei to. Nothing in Madei's world rewards a mark. It doesn't feed anyone, warm anyone, or carry anything. Madei just made it, and walked away.

The same year, someone named Miyomis tied two pieces of wood together, and within a year people were floating across open water to land nobody could walk to. A year later, someone named Tabek buried seed on purpose, after seeing seed come back as food. And by the second year, two people named Dures and Niyadi, who were not related in any way, had been passing food back and forth for years, and kept on doing it. In the worlds in this book, food mostly stayed inside families. Within four years, there were four partnerships like that one, between strangers.

I don't know what any of it will become. Maybe the partnerships will fade, the way the planners' lessons did. Maybe the buried seed will be forgotten, like the raft and the fired clay. Maybe one of those strangers will teach a child to do the same.

And the mark may still be there, on the rock. People walk past it. Every week, someone might look at it and, for the first time, think it means something: a place, a person, a warning, a name. If that ever happens, it will be the first word in that world, and nobody will have written it.

I'll be watching when it does.

That is where Part Two begins.

Sources

THE RESEARCH THIS book mentions, for anyone who wants to read further. Every result about my own worlds comes from the project's records, which are public at github.com/abu4evere/simulated-world.

Artificial worlds and agents

- Ray, T. S. (1991). An approach to the synthesis of life. In *Artificial Life II*. Addison-Wesley. (Tierra.)
- Ofria, C. and Wilke, C. O. (2004). Avida: a software platform for research in computational evolutionary biology. *Artificial Life* 10(2).
- Epstein, J. M. and Axtell, R. (1996). *Growing Artificial Societies: Social Science from the Bottom Up.* MIT Press. (Sugarscape.)
- Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R. and others (2023). Generative agents: interactive simulacra of human behavior. arXiv:2304.03442.
- Altera.AL and others (2024). Project Sid: many-agent simulations toward AI civilization. arXiv:2411.00114.

How culture grows and is lost

- Henrich, J. (2004). Demography and cultural evolution: how adaptive cultural processes can produce maladaptive losses. The Tasmanian case. *American Antiquity* 69(2).
- Henrich, J. and Gil-White, F. J. (2001). The evolution of prestige. *Evolution and Human Behavior* 22(3).
- Kline, M. A. and Boyd, R. (2010). Population size predicts technological complexity in Oceania. *Proceedings of the Royal Society B* 277.
- Derex, M., Beugin, M.-P., Godelle, B. and Raymond, M. (2013). Experimental evidence for the influence of group size on cultural complexity. *Nature* 503.

How foragers live, move and store

- Woodburn, J. (1982). Egalitarian societies. *Man* 17.
- Binford, L. R. (1980). Willow smoke and dogs' tails: hunter-gatherer settlement systems and archaeological site formation. *American Antiquity* 45(1).
- Testart, A. (1982). The significance of food storage among hunter-gatherers. *Current Anthropology* 23(5).
- Kelly, R. L. (2013). *The Lifeways of Hunter-Gatherers: The Foraging Spectrum.* Cambridge University Press.
- Hamilton, M. J., Milne, B. T., Walker, R. S., Burger, O. and Brown, J. H. (2007). The complex structure of hunter-gatherer social networks. *Proceedings of the Royal Society B*.
- White, A. (2017). A model-based analysis of the minimum size of demographically-viable hunter-gatherer populations. *Journal of Artificial Societies and Social Simulation* 20(4).
- Hill, K. R., Wood, B. M., Baggio, J., Hurtado, A. M. and Boyd, R. T. (2014). Hunter-gatherer inter-band interaction rates. *PLOS ONE* 9(7).
- Dyble, M., Migliano, A. B., Page, A. E. and Smith, D. (2021). Relatedness within and between Agta residential groups. *Evolutionary Human Sciences*.
- Marlowe, F. W. (2010). *The Hadza: Hunter-Gatherers of Tanzania.* University of California Press.
- Padilla-Iglesias, C. and others (2025). Sexual division of labour shapes hunter-gatherer spatial ranges. bioRxiv (preprint).
- Venkataraman, V. V., Kraft, T. S., Dominy, N. J. and Endicott, K. M. (2017). Hunter-gatherer residential mobility and the marginal value of rainforest patches. *PNAS* 114.

Hunger, the body and hard years

- Wells, J. C. K. (2006). The evolution of human fatness and susceptibility to obesity: an ethological approach. *Biological Reviews* 81(2).

- Simondon, K. B. and others (2004). Season of birth is not associated with risk of early adult death in rural Senegal. *International Journal of Epidemiology* 33(1).
- Townsend, R. M. (1994). Risk and insurance in village India. *Econometrica* 62(3).

The bigger questions

- Bostrom, N. (2003). Are you living in a computer simulation? *Philosophical Quarterly* 53(211).
- The Qur'an. Chapters cited: Al-Fatiha, Al-Baqarah, Al-Imran, Al-Anbiya, Fatir, Ash-Shura, Adh-Dhariyat, Al-Qamar, Al-Mulk, Al-Alaq.
- Everett, H. (1957). "Relative state" formulation of quantum mechanics. *Reviews of Modern Physics* 29(3).